How Much Can a Temperature Log Tell Us About Spoiling Milk?

A Loss-Budget Benchmark for Cold-Chain Monitoring

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ABSTRACT

A cold-chain monitor records temperature, but bacterial growth in milk also depends on strain, starting load and lag, none of which a thermometer observes. This paper measures how much of the error in predicting milk spoilage from a temperature record is caused by unobserved biology, by the sensor, by the sampling interval and by the choice of decision rule. An open Monte Carlo benchmark of 240 000 simulated trips is built in which a Baranyi–Roberts model with a square-root temperature law stands in for real milk, and a Gompertz shelf-life accumulator that runs on a microcontroller predicts it from noisy, sampled records. Rules are compared by the area under the receiver operating characteristic curve (AUC) with a paired bootstrap on identical trips, against an oracle that knows each trip's biology. The model reproduces a published fluctuating-temperature experiment within 4 %, fits 50 ComBase records of natural spoilage flora, and predicts 53 held-out *Listeria monocytogenes* growth rates with an accuracy factor of 1.37. Unobserved biology costs 0.27–0.40 AUC; a platinum resistance sensor instead of a digital one gains at most 0.02; hourly sampling is indistinguishable from minute-by-minute sampling in 11 of 12 settings; and the kinetic score improves on a plain degree-hour count by at most 0.04. A Monte Carlo probability is far better calibrated than a single alarm (expected calibration error 0.03–0.07 against 0.34–0.43). For this problem, accuracy is limited by biology, not by the thermometer.

1 Introduction

Milk is perishable in a precise way. Psychrotrophic bacteria that grow slowly at 4 °C grow several times faster at 13 °C, and every hand-off between dairy, truck, depot, shop and home is an opportunity for the temperature to drift. Predictive microbiology describes this with a primary model for the bacterial count against time, a secondary model for the growth rate against temperature, and a tertiary model that applies both along a temperature history [1–3]. Temperature loggers are now cheap enough to travel with every crate [4–6], and dynamic shelf-life labels that replace a printed date by a prediction from the log have been proposed for dairy and meat [7–9]. The question this paper answers is the one such a device faces: *given only a temperature record, how well can an inexpensive calculation tell whether milk has passed a limit, and what prevents it from doing better?*

The stakes are economic and sanitary. Post-pasteurisation contamination with psychrotrophic bacteria causes roughly half to three-fifths of fluid-milk spoilage in the United States [10, 11]. Foodborne disease causes an estimated 600 million illnesses and 420 000 deaths a year, 175 000 of them in the WHO South-East Asia region [12]; these deaths are dominated by agents other than milk-borne bacteria, and the figures are quoted as context only. The limits a monitor reports against are process-hygiene criteria: the Food Safety and Standards Authority of India (FSSAI) sets an aerobic plate count of 3×10^4 CFU mL^{-1} for pasteurised milk [13], which is adopted here as the first limit. A recent review of fresh dairy concludes that few studies predict the end of shelf life, that few models are validated beyond their development data, and that probabilities are preferable to single verdicts [14].

1.1 Prior work

Growth models are fitted under constant conditions and applied to changing ones [1,15]. The modified Gompertz curve [16,17] and the Baranyi–Roberts model [18, 19] are the standard primary models; the square-root law [20] and the cardinal-temperature model [21] are standard secondary models; and ComBase gathers growth records to fit them [22, 23]. For psychrotrophic *Pseudomonas* in milk, the square-root law was fitted to 68 growth curves and tested in industry [24, 25], and growth under fluctuating temperature has been measured directly [26, 27].

The closest studies are Monte Carlo analyses of shelf life [28]. For fluid milk, Lau et al. modelled spoilage from Gram-negative post-pasteurisation contamination and found the maximum growth rate and storage temperature most influential [29]; Buehler et al. and Qian et al. modelled psychrotolerant spore-formers and deployed the model as a supply-chain tool, with home storage temperature, transport temperature and initial spore load ranked highest [30, 31]. Giannakourou and Taoukis propagated kinetic uncertainty and cold-chain variability through nested Monte Carlo loops [32]. On the monitoring side, time–temperature indicators integrate an Arrhenius rate with a fixed activation energy [33]; the Safety Monitoring and Assurance System predicts remaining shelf life from logged temperature [7, 34]; field studies reveal large gaps between dynamic and printed shelf life [8]; mean kinetic temperature has been used for milk and chicken [9]; neural networks estimate product temperature from vehicle data [35]; heat-transfer simulation quantifies the cost of cold-chain failures [36, 37]; and a recent framework couples a growth model with learned components for milk [38]. Table 1 sets these approaches side by side.

These studies ask how variable the outcome is, which input moves it most, or how to build a decision system. None asks

Table 1
Approaches to predicting spoilage from a temperature history, and the question each leaves open for a temperature monitor

Approach	Input	Temperature law	Biological variability	Output	Open question for a monitor
Threshold alarm, time above a limit	logged temperature	step	ignored	alarm	how often the alarm is wrong
Degree-hours	logged temperature	linear	ignored	index	how close it comes to a growth model
TTI, mean kinetic temperature [9, 33]	logged or physical integral	Arrhenius, fixed E_a	ignored	equivalent temperature	accuracy lost to biology
Dynamic shelf life, SMAS [7, 34]	logged temperature	square-root	nominal	remaining shelf life	loss to biology, sensor and sampling
Monte Carlo spoilage models [29–31]	simulated temperature	square-root	sampled	share of units spoiled	what remains once temperature is logged
Nested uncertainty [32]	simulated temperature	Arrhenius	sampled	shelf-life distribution	accuracy of a monitor
Product-temperature estimation [35]	vehicle and door data	Q_{10}	ignored	product temperature	effect on ranking of spoiled units
Decision intelligence [38]	sensor stream	growth model + learning	partly	decision	decomposition of the error
This work	**sensed, sampled log**	**square-root**	**sampled, with oracle**	**AUC loss budget, probability**	–

the question a monitor faces: starting from a *measured* record and judged against a known truth, how much of the ability to rank trips by spoilage is lost to unobserved biology, to sensor error, to the sampling interval and to the choice of rule? The two views are complementary. When temperature is unknown it dominates the spread of outcomes, as the sensitivity analyses find; once it is logged, what remains is mostly biology, which is what a loss budget measures.

1.2 An illustrative trip and the aim of the study

Figure 1 shows the problem on a single trip. Three cartons travel together and therefore share one temperature history; they differ only in their starting load and the physiological state of their bacteria. A monitor that reads the log returns one answer for all three. It is late for one carton and early for the other two, and no improvement of the thermometer can change that, because the information it lacks is not in the temperature record. The aim of this study is to quantify that limit and to rank it against the losses a designer can control. Four research questions are addressed: (RQ1) how much ranking accuracy is lost to unobserved biology, to the sensor and to sampling; (RQ2) how much a better sensor or faster sampling recovers; (RQ3) how a growth-model score compares with simple rules on identical trips; and (RQ4) whether a probability is better calibrated than a single alarm.

2 Models

2.1 Secondary model

The square-root (Ratkowsky) law relates the maximum specific growth rate μ (h^{-1}, natural-log basis) to temperature below the optimum [20],

$$\sqrt{\mu(T)} = b\,(T - T_{\min}), \qquad T > T_{\min}. \tag{1}$$

Neumeyer fitted it to 68 growth curves of a fast-growing *Pseudomonas* in broth and milk, giving $T_{\min} = -7.7\,^{\circ}\mathrm{C}$ and $b = 0.030\,\mathrm{h}^{-1/2}\,\mathrm{K}^{-1}$ [24]. Rates of *P. fluorescens* in UHT milk at 4, 15 and 29 °C [26] give $b = 0.027$ and $T_{\min} = -9.0\,^{\circ}\mathrm{C}$; the two sources agree within 15 % on generation time at 4 °C (5.6 and 6.4 h). Equation (1) implies $\mu(13)/\mu(4) = [(13 - T_{\min})/(4 - T_{\min})]^2 = 3.1$ for $T_{\min} = -7.7\,^{\circ}\mathrm{C}$: one hour at 13 °C is worth about three hours at 4 °C.

2.2 Reference model

The Baranyi–Roberts model [18, 19], which stands in for real milk, describes the log population $y = \ln N$ and a physiological state q:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \mu(T)\,\frac{q}{1+q}\left(1 - \mathrm{e}^{y - y_{\max}}\right), \tag{2}$$

$$\frac{\mathrm{d}q}{\mathrm{d}t} = \mu(T)\,q, \qquad h_0 = \ln(1 + 1/q_0)\,. \tag{3}$$

At constant temperature the lag is $\lambda = h_0/\mu$, so the product $\lambda\mu = h_0$ is constant for cells with the same history. The model handles changing temperature naturally, because μ is evaluated along the trace.

2.3 Edge estimator

The monitor itself uses a different primary model, so that the comparison is not circular. Let $\mathrm{SL}(T)$ be the time for a population at constant T to reach a limit N_L under the modified Gompertz curve [17],

$$\mathrm{SL}(T) = \frac{h_0}{\mu(T)} + \frac{A}{\mu_{10}(T)\,\mathrm{e}}\left[1 - \ln\left(-\ln\frac{L}{A}\right)\right], \tag{4}$$

with $\mu_{10} = \mu/\ln 10$, $A = \log_{10}(N_{\max}/N_0)$ and $L = \log_{10}(N_L/N_0)$. A reading T_i held over an interval Δt_i consumes a fraction $\Delta t_i/\mathrm{SL}(T_i)$ of the shelf life, and the cumulative shelf life is

$$\mathrm{CSL} = \sum_i \frac{\Delta t_i}{\mathrm{SL}(T_i)}, \qquad \text{limit reached when } \mathrm{CSL} \geq 1. \tag{5}$$

Because every term of Eq. (4) is proportional to $1/\mu$, $\mathrm{SL}(T) = K/\mu(T)$ with a constant K that depends only on h_0, N_0, N_L and $N_{\max}$, and therefore

$$\mathrm{CSL} = \frac{1}{K}\sum_i \mu(T_i)\,\Delta t_i \;\approx\; \frac{1}{K}\int_0^t \mu\big(T(s)\big)\,\mathrm{d}s. \tag{6}$$

The monitor thus integrates the growth rate along the log and compares the result with a fixed budget K. The temperature

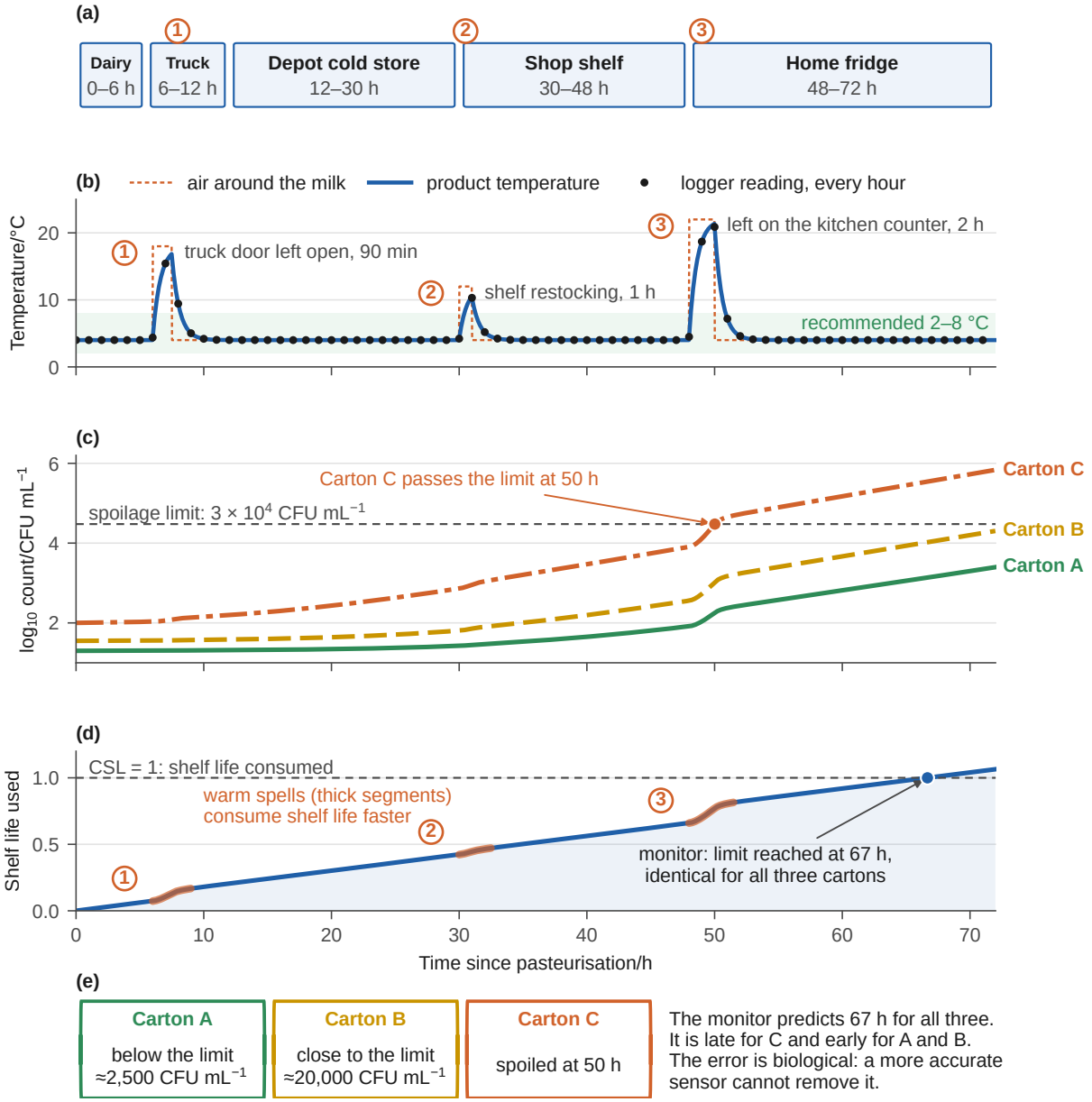


Fig. 1 One delivery, three cartons. (a) Stages of a 72-h trip. (b) Ambient air and product temperature with three excursions, and the hourly logger readings. (c) Simulated counts in three cartons that share the same history but differ in starting load and lag. (d) Cumulative shelf life (CSL) computed by the monitor from the log; warm spells consume shelf life faster. (e) Outcome: one prediction for three cartons whose true fates differ

record determines the integral; the budget, which carries the biology, is not observed. Equation (5) costs one table lookup and one addition per reading and runs on a microcontroller.

3 Benchmark

Figure 2 summarises the design: a simulated world supplies the truth, and a monitor that sees only a sensed, sampled record tries to recover it.

3.1 Trips and labels

Each trip has a product-temperature trace on a one-minute grid. The base temperature is drawn from 2.5–5 °C. Excursions arrive at a mean rate of one per 144 h, peak at 12, 14.3, 17.5 or 30.7 °C (weights 0.30, 0.25, 0.30, 0.15) and last a lognormal time with median 60 min, clipped to 10–480 min. Product temperature follows ambient with a first-order time constant of 5–20 min for a low-mass pouch and 1–4 h for an insulated crate. A reference population grows along the trace by Eqs. (2)–(3), with $T_{\min} \sim \mathcal{N}(\bar{T}_{\min}, \sigma_T^2)$, a lognormal multiplier on b with coefficient of variation $cv \in \{0, 0.10, 0.25\}$, h_0 uniform over a measured range and a random starting load.

A trip is *spoiled* if the reference population exceeds the limit within the label horizon, defined as the time the estimator takes to reach the limit at a constant 4 °C with nominal parameters, rounded up to a whole hour, as a printed shelf life would be. The horizon was fixed from label balance before any rule was compared. Prevalence is therefore constructed (about 0.5 for spoilage, 0.08–0.38 for *Listeria*), and absolute predictive values must not be read as field rates.

3.2 Sensors, sampling and scores

Sensors add a per-trip offset, white noise and quantisation: a DS18B20 digital sensor (offset within ±0.5 °C, noise 0.05 °C, step 0.0625 °C) [39], a PT1000 platinum resistance sensor read by a MAX31865 (±0.15 °C, 0.03 °C, 0.03125 °C) [40], and an ideal sensor. Sampling intervals are 1, 15 and 60 min, plus an adaptive rule that samples every minute while the temperature changes and every 15 min otherwise. Five scores

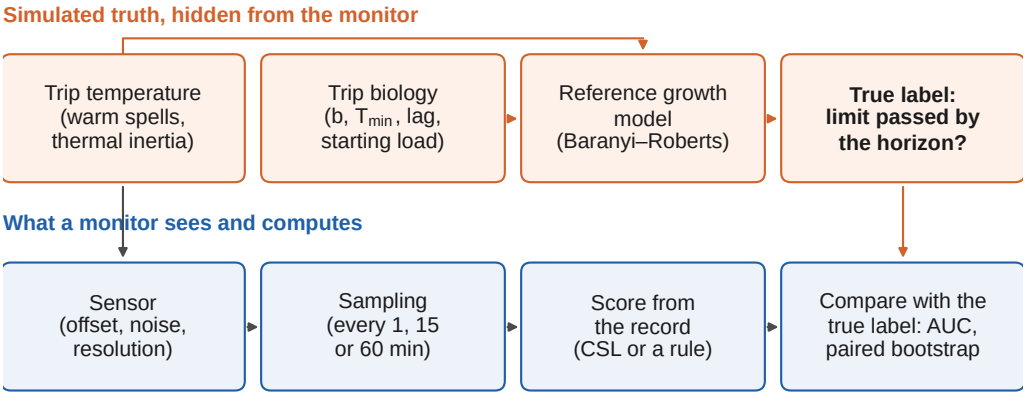


Fig. 2 Design of the benchmark. Each simulated trip has a true temperature history and a true biology, which the reference model turns into a true label (top). The monitor sees only a sensed, sampled record and turns it into a score (bottom). Scores are judged against the true labels by AUC; removing information step by step gives the loss budget

Table 2
Benchmark runs. "Assumed" marks values without a source

Run	Biology and labels
Baseline	$b = 0.030$, $T_{\min} = -7.7\,°C$ [24]; $\sigma_T = 1.5\,K$ (assumed); $h_0 \in [2.5, 5]$ [26]; N_0 30–100 CFU mL^{-1} (assumed); limits 3×10^4 and 10^6 CFU mL^{-1}; horizons 83 and 113 h
Extra lag	as baseline, plus extra lag after shifts [27]
Low load	as baseline, N_0 0.1–3 CFU mL^{-1} [10]; horizons 120 and 154 h
Milk flora	$b = 0.0256$, $T_{\min} = -6.6\,°C$ (ComBase, $n = 50$), low load; horizons 201 and 258 h
Listeria	$b = 0.0235$, $T_{\min} = -2.3\,°C$ (ComBase, $n = 46$); $h_0 \in [2, 4.5]$, $\sigma_T = 1.0\,K$, N_0 0.06–0.18 CFU mL^{-1} (assumed); labels +0.5 and +1 log; horizons 100 and 202 h

are computed from the measured record: CSL; degree-hours above 4 °C; time above 5 °C; time above 8 °C; and maximum temperature. An *oracle* CSL uses each trip's true parameters and noise-free temperature and sets the ceiling. In a structure-risk run, a quarter of the lag state is re-imposed whenever the product temperature departs from its two-hour running mean by more than 4 °C [27]; the estimator is not told about it.

3.3 Parameter sets

Table 2 lists the five runs. The baseline uses the Neumeyer parameters and a measured lag range. Because parameters from inoculated isolates need not describe natural flora, a fourth run uses parameters fitted to ComBase records of spoilage flora (Sect. 4), and a fifth models *Listeria monocytogenes* with labels of +0.5 and +1 $\log_{10}$ growth above the starting load (round-number labels, not regulatory limits). Each run has 12 settings (two limits, two products, three values of cv) of 4 000 trips, so the benchmark contains 240 000 trips, each read with every sensor, sampling rule and score.

3.4 Statistical analysis

The primary metric is the area under the ROC curve, $\mathrm{AUC} = \Pr(S_1 > S_0) + \frac{1}{2}\Pr(S_1 = S_0)$, the probability that a randomly chosen spoiled trip scores higher than a randomly chosen unspoiled one [41]. It depends on ranking, not on a threshold or on the constructed prevalence, and is estimated by the Mann–Whitney statistic [42]. Its standard error for $A = 0.67$ and $n_1 = n_0 = 2\,000$ is about 0.0085 by the Hanley–McNeil formula [41], which fixed the size of each setting: the biology gap is more than fifteen standard errors, while sensor and sampling effects are of the order of one and can be resolved only by paired comparison.

Two scores computed on the same trips are strongly correlated, so their difference $\hat{D} = \widehat{\mathrm{AUC}}_1 - \widehat{\mathrm{AUC}}_2$ has variance $\mathrm{Var}(\widehat{\mathrm{AUC}}_1) + \mathrm{Var}(\widehat{\mathrm{AUC}}_2) - 2\,\mathrm{Cov}(\widehat{\mathrm{AUC}}_1, \widehat{\mathrm{AUC}}_2)$, which is much smaller than the sum of the separate variances. Every trip is therefore read with every sensor, sampling rule and score (a common-random-numbers design), and intervals for differences come from a paired bootstrap of trips with $B = 400$ resamples [43], the resampling counterpart of the method of DeLong et al. [44]; single AUCs use $B = 200$. No conclusion is drawn from a single setting: for each contrast the range over settings is reported, together with the number of settings whose 95 % interval excludes zero.

An uncertainty-aware variant draws $M = 80$ parameter sets from a prior, computes CSL under each and reports the fraction that reach the limit as a probability. It is scored against a single alarm with nominal parameters by the Brier score [45] and by the expected calibration error over ten equal-width bins [46], $\mathrm{ECE} = \sum_m (|B_m|/n)\,|\bar{y}_{B_m} - \bar{p}_{B_m}|$.

4 Validation against real data

4.1 A fluctuating-temperature experiment

Lin et al. held UHT milk inoculated with *P. fluorescens* at 4 °C with two four-hour excursions to 29 °C, and report that the time to 10^6 CFU mL^{-1} fell from about 90 h to about 55 h [26]. With parameters fitted only to their growth rates and their 4 °C lag, the reference model gives 90 h and 53 h, an error of −4 % on the excursion case, which was not used for fitting. This is one experiment with one organism and an inoculated load far above real post-pasteurisation levels.

4.2 Natural spoilage flora

The 51 ComBase records of aerobic spoilage bacteria in pasteurised milk span 2.7–14.9 °C [23]. One record (6.8 °C) lies far below its neighbours; without it the square-root law fits the remaining 50 records with $b = 0.0256$, $T_{\min} = -6.6\,°C$ and a residual standard deviation of 0.16 (natural-log units), and with it $b = 0.0260$ and $T_{\min} = -6.3\,°C$, so the conclusions do not depend on it. Parameters from inoculated isolates overpredict these records with a bias factor [47] of 1.62 that is nearly constant across temperature (Fig. 3): the temperature *shape* transfers from isolates to natural flora, the *level* does not.

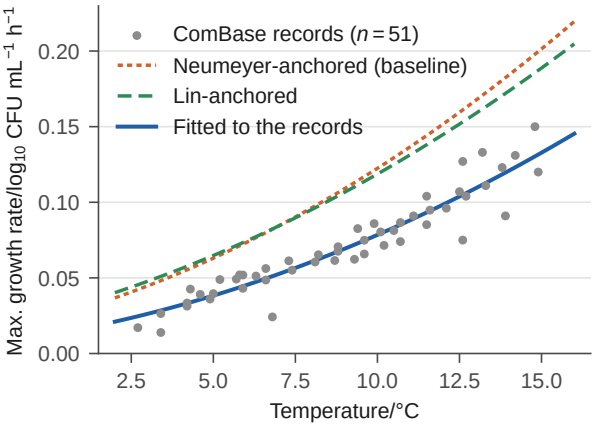


Fig. 3 Maximum growth rates of natural spoilage flora in pasteurised milk (points; ComBase [23]) with the Neumeyer-anchored model (dotted), the Lin-anchored model (dashed) and the square-root fit to the records (solid)

Table 3
Composition of the compiled ComBase milk database by organism. *n*: records; Rate: maximum growth rate stated; Curve: growth curve only; None: no growth reported; N/T: rate not transcribed

Organism	*n*	Rate	Curve	None	N/T
L. monocytogenes/innocua	149	102	43	0	4
Bacillus cereus	94	47	36	3	8
Escherichia coli	66	22	15	4	25
Salmonella spp.	25	4	7	13	1
Campylobacteraceae	16	0	13	0	3
Shigellae	11	0	10	0	1
Cronobacter spp.	6	0	5	0	1
Aerobic spoilage bacteria	4	4	0	0	0
Lactic acid bacteria	1	0	1	0	0
Other	1	0	1	0	0
Total	373	179	131	20	43

4.3 *Listeria monocytogenes*

ComBase milk records with a stated rate for *L. monocytogenes* at 2–16 °C were split by condition: 46 records without added factors were used for fitting and 53 records with other stated conditions (sterilised or previously heated milk, added salts or acids) were held out. The fit gives $b = 0.0235$ (95 % bootstrap interval 0.0203–0.0265) and $T_{\min} = -2.3$ °C (−4.2 to −0.9 °C), a generation time of 32 h at 4 °C, and on the held-out records a bias factor of 1.02 and an accuracy factor of 1.37 (1.33 on the fitted records). Because $T_{\min}$ lies closer to refrigeration temperature, the rate at 13 °C is 5.9 times that at 4 °C for *Listeria*, against about 3.4 times for spoilage flora (Fig. 4). The split is by condition rather than by study, so the test is weaker than a split by source.

4.4 The compiled data

ComBase returned about 480 records with milk as the food matrix, of which 373 were compiled (Table 3), together with the 51 records of spoilage flora. All compiled records are released with the code.

Table 4
Loss budget: mean AUC over the 12 settings of each run as information is removed, and the range of the biology gap (oracle minus ideal sensor)

Run	Oracle	Ideal	PT1000	DS18B20	DS18B20	Biology gap
		1 min	15 min	15 min	60 min	
Baseline	1.000	0.671	0.670	0.661	0.662	0.27–0.40
Extra lag	0.988	0.652	0.652	0.644	0.645	0.27–0.40
Low load	1.000	0.667	0.666	0.656	0.657	0.28–0.40
Milk flora	1.000	0.664	0.663	0.654	0.653	0.28–0.40
Listeria	0.950	0.738	0.737	0.725	0.726	0.09–0.34

5 Results

Principal findings

- Unobserved biology costs 0.27–0.40 AUC in the baseline run, an order of magnitude more than any sensor or sampling choice.
- A PT1000 instead of a DS18B20 gains at most 0.02 AUC; hourly sampling matches one-minute sampling in 11 of 12 settings.
- The kinetic score beats time above 5 °C and maximum temperature in every setting, but improves on degree-hours by at most 0.04.
- A Monte Carlo probability reaches ECE 0.03–0.07; a single alarm 0.34–0.43.

5.1 Loss budget (RQ1)

An oracle that knows each trip's biology and temperature separates spoiled from unspoiled trips almost perfectly, by construction. The same CSL with nominal biology and an ideal sensor loses 0.27–0.40 AUC across the 12 baseline settings (Table 4, Fig. 5). The loss is large even when strains do not differ in growth rate ($cv = 0$), because the minimum growth temperature, lag state and starting load still differ between trips. In terms of Eq. (6), these quantities set the budget K and the rate scale, which the temperature record does not contain; no sensor or sampling rate can recover them. Replacing the ideal sensor by a PT1000 or a DS18B20 sampled every 15 or 60 min removes only a further 0.00–0.02.

5.2 Sensors and sampling (RQ2)

Replacing a DS18B20 with a PT1000 gains 0.00–0.02 AUC at 15-min sampling (Table 5). A sensor offset of ±0.5 °C shifts $\sqrt{\mu}$ by $b \times 0.5$, about 4 % of its value at 4 °C, whereas the trip-to-trip spread of $T_{\min}$, h_0 and N_0 moves the shelf-life budget by tens of per cent; the ordering of the two losses follows directly. Sampling hourly instead of every minute shows no detectable difference in 11 of 12 settings and at most about 0.01 where it does, and the adaptive rule is indistinguishable from fixed one-minute sampling. These results depend on the assumed excursion model (minimum duration 10 min, median 60 min) and would change if short, sharp excursions were common.

5.3 Against simple rules (RQ3)

On identical trips, the kinetic score beats time above 5 °C in 12 of 12 settings and maximum temperature in 12 (Table 6). Against degree-hours above 4 °C, a rule that needs no growth model, the gain is positive with an interval excluding zero in 10 of 12 settings but small: 0.00–0.02 AUC in the baseline run and up to 0.04 when extra lag is present. This is expected: near refrigeration temperature, Eq. (1) is close to linear in T, so a degree-hour count is a first-order approximation of the

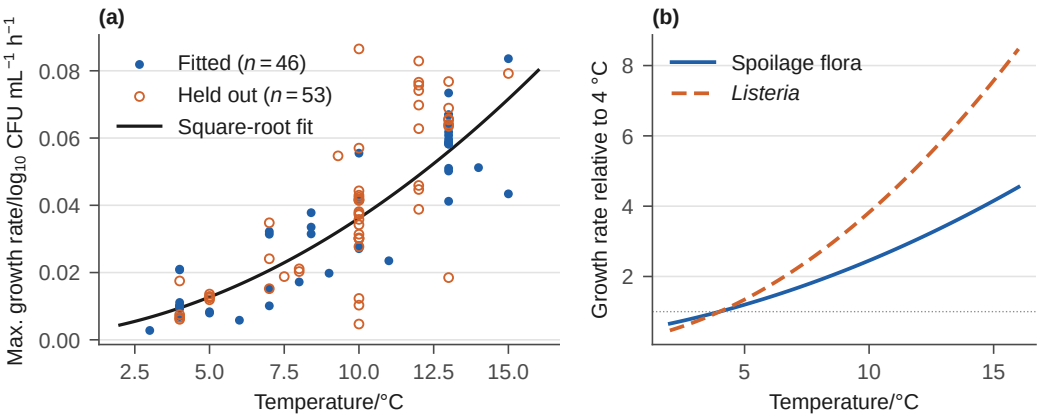


Fig. 4 *Listeria monocytogenes* in milk. (a) Maximum growth rates used for fitting (filled) and held out by condition (open), with the square-root fit (ComBase [23]). (b) Growth rate relative to 4 °C for spoilage flora (solid) and *Listeria* (dashed): a warm hour costs more for the pathogen

Table 5
Paired AUC differences on identical trips: minimum/maximum over the 12 settings of each run, with the number of settings (of 12) whose 95 % paired-bootstrap interval excludes zero in brackets

Contrast	Baseline	Extra lag	Low load	Milk flora	*Listeria*
CSL − degree-h	0.001/0.023 (10)	0.006/0.037 (12)	0.001/0.023 (8)	0.004/0.025 (10)	0.002/0.023 (8)
CSL − time > 5 °C	0.008/0.074 (12)	0.001/0.104 (11)	0.005/0.066 (11)	0.006/0.075 (11)	0.010/0.077 (12)
PT1000 − DS18B20	0.003/0.015 (9)	0.003/0.013 (7)	0.005/0.018 (10)	0.002/0.015 (8)	0.005/0.020 (8)
60 min − 1 min	-0.008/0.004 (1)	-0.008/0.004 (1)	-0.011/0.008 (2)	-0.009/0.006 (0)	-0.008/0.009 (0)
Adaptive − 1 min	-0.006/0.005 (0)	-0.006/0.006 (0)	-0.008/0.009 (1)	-0.007/0.006 (0)	-0.006/0.009 (0)

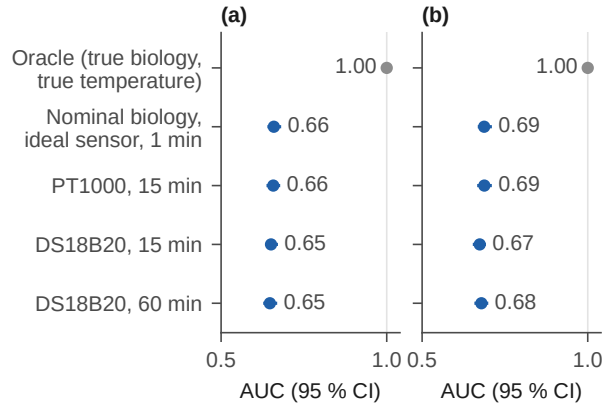


Fig. 5 Loss budget for (a) a low-mass pouch and (b) an insulated crate (limit 3×10^4 CFU mL^{-1}, $cv = 0.10$). Grey: oracle with true biology and temperature. Blue: the kinetic score as information is removed step by step. Bars: 95 % bootstrap intervals

integral in Eq. (6). Most of the value of the kinetic score is therefore available from a degree-hour count; its advantage lies in the larger excursions, where the rate is strongly curved, and in its biological interpretation.

5.4 Robustness across parameter sets

Figure 6 summarises the four effects across all five runs. The pattern holds whether the biology is anchored to inoculated isolates, includes extra lag, starts from a realistic low load, is fitted to natural flora or describes *Listeria*: the biology gap is 0.28–0.40 (low load), 0.28–0.40 (milk flora) and 0.09–0.34 (*Listeria*); the gain over degree-hours stays below 0.04 in every run; and the sensor gain stays small (0.01–0.02 for *Listeria*).

5.5 Calibration (RQ4)

A single alarm with nominal parameters is badly calibrated against the constructed labels (ECE 0.34–0.43; Table 7). A Monte Carlo probability with a prior that matches the truth reaches ECE 0.03–0.07, and one with an overconfident prior on the growth rate 0.04–0.13. Absolute values reflect the constructed prevalence; the ordering of the three variants is the finding. The *Listeria* run requires a caveat. With the +1 log label the probability reaches ECE 0.08–0.20 against 0.34–0.41 for the alarm. With the half-log label the limit falls inside the lag-to-growth transition, where the Gompertz estimator approximates the reference poorly; even the oracle reaches only 0.88–0.92 AUC there, and the probability, although still better than the alarm (ECE 0.28–0.41 against 0.42–0.45), is poorly calibrated in absolute terms. The choice of label interacts with the structure of the estimator.

6 Monitor architecture

Figure 7 shows how the estimator runs on a logger. A probe reading is taken on an adaptive schedule, the shelf-life rate $1/\mathrm{SL}(T)$ is read from a stored table, Eq. (5) is updated, and the same sum is carried for 80 sampled biological parameter sets to produce a probability. Parameter fitting is done once, offline. A browser demonstrator runs the same kernel (https://copyninja07-07.github.io/dairyguard/); a parity test ensures that it reproduces the Python implementation. A hardware logger is future work.

7 Discussion

7.1 Where effort belongs

In every run, what the thermometer cannot observe costs about an order of magnitude more accuracy than any choice of sensor or sampling interval. Equation (6) explains why: a log measures the integral of the growth rate well, but the budget against which the integral is compared varies from carton to carton. A more accurate thermometer is therefore a poor way to improve a spoilage estimate, and a low-cost digital probe sampled every 15–60 min loses little under the assumptions made here. Effort is better spent on the biology: measuring

Table 6
AUC of the five scores from a DS18B20 record sampled every 15 min (baseline run, 4 000 trips per row). The best score in each row is in bold. The last two columns give the same kinetic score with an ideal sensor sampled every minute, and the oracle that knows each trip's biology

Limit	Product	*cv*	CSL	Degree-h > 4 °C	Time > 5 °C	Time > 8 °C	T_{max}	Ideal, 1 min	Oracle
A	Pouch	0.00	**0.705**	0.690	0.635	0.569	0.629	0.718	1.000
A	Pouch	0.10	**0.651**	0.634	0.599	0.557	0.591	0.659	1.000
A	Pouch	0.25	**0.620**	0.619	0.591	0.561	0.602	0.627	1.000
A	Crate	0.00	**0.722**	0.711	0.701	0.586	0.638	0.732	1.000
A	Crate	0.10	**0.674**	0.666	0.657	0.577	0.611	0.687	1.000
A	Crate	0.25	**0.599**	0.596	0.592	0.541	0.571	0.603	1.000
B	Pouch	0.00	**0.710**	0.686	0.636	0.564	0.587	0.723	1.000
B	Pouch	0.10	**0.656**	0.634	0.606	0.539	0.564	0.670	1.000
B	Pouch	0.25	**0.604**	0.594	0.565	0.531	0.549	0.608	1.000
B	Crate	0.00	**0.708**	0.696	0.687	0.600	0.629	0.724	1.000
B	Crate	0.10	**0.676**	0.670	0.663	0.585	0.611	0.686	1.000
B	Crate	0.25	**0.603**	0.595	0.591	0.538	0.552	0.608	1.000

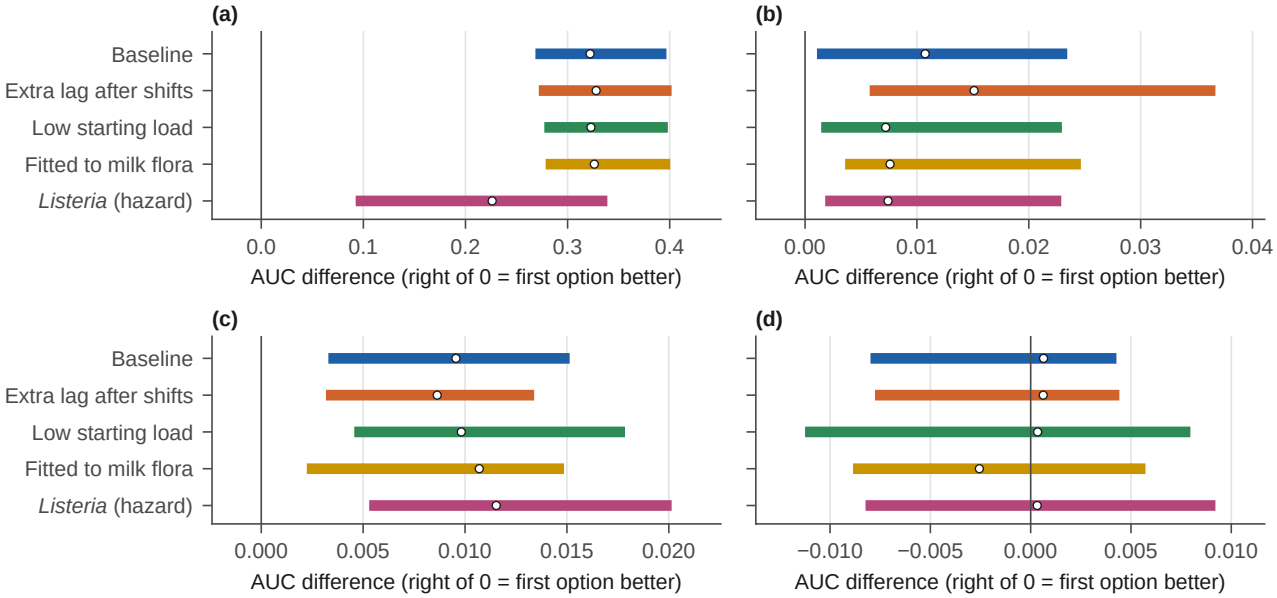


Fig. 6 Four effects across the five parameter sets, as AUC differences on identical trips: (a) loss to unobserved biology, (b) kinetic score over degree-hours, (c) PT1000 over DS18B20, (d) hourly over one-minute sampling. Bars span the 12 settings of each run; dots mark the median. Horizontal scales differ

the starting load or a proxy of handling history, learning the lag and strain spread of the product at hand, and reporting a probability instead of a verdict.

7.2 Consistency with prior findings

Sensitivity analyses rank temperature first [29–32]; this study finds that biology dominates. The two results answer different questions. When temperature is drawn from a distribution, it drives the spread of outcomes; once it is measured, the residual uncertainty is biological. The small gain of the kinetic score over degree-hours agrees with the long practical use of degree-hour and time–temperature integrators [9, 33].

7.3 Hazard and spoilage

For spoilage flora a warm hour is costly but the baseline rate is high; for *Listeria*, growth at 4 °C is slow (generation time 32 h) but rises about 5.9-fold by 13 °C. The same benchmark and the same design conclusions hold for both. This paper does not estimate illness risk, which depends on dose, virulence, host and the contamination event, none of which are modelled.

7.4 Limitations

(i) The reference and the estimator both reduce the temperature history to an integral of the growth rate, so the benchmark measures how much accuracy is lost to the listed sources, not whether a kinetic score models growth better; the structure-risk run relaxes one structural assumption only. (ii) Strain spread, starting load, excursion statistics and the *Listeria* lag are assumed. (iii) Labels are constructed, with prevalence near 0.5 for spoilage and 0.08–0.38 for *Listeria*. (iv) ComBase records come from heterogeneous studies and report maximum rates, not curves, so lag and responses to shifts are untested on them. (v) A single experiment validates fluctuating temperature, and no new plate counts were made. (vi) Neither organism class represents spore-formers, which account for a large share of spoilage [10].

8 Conclusions

A temperature log can say much about how much growth the milk *could* have had, and little about how much it *did*. In an open benchmark of 240 000 trips anchored to published and ComBase data, unobserved biology costs 0.27–0.40 AUC; a degree-hour count comes within 0.04 of a growth model;

Table 7
Calibration: range of expected calibration error over the 12 settings, and mean Brier score, for each run. Lower is better

Variant	Baseline	Extra lag	Low load	Milk flora	*Listeria*
Probability, matched prior	0.03–0.07 / 0.231	0.01–0.07 / 0.236	0.02–0.11 / 0.233	0.02–0.09 / 0.234	0.08–0.41 / 0.216
Probability, overconfident	0.04–0.13 / 0.237	0.05–0.13 / 0.244	0.02–0.14 / 0.238	0.02–0.13 / 0.239	0.11–0.44 / 0.218
Single alarm	0.33–0.43 / 0.380	0.34–0.44 / 0.391	0.34–0.44 / 0.386	0.34–0.43 / 0.385	0.34–0.45 / 0.400

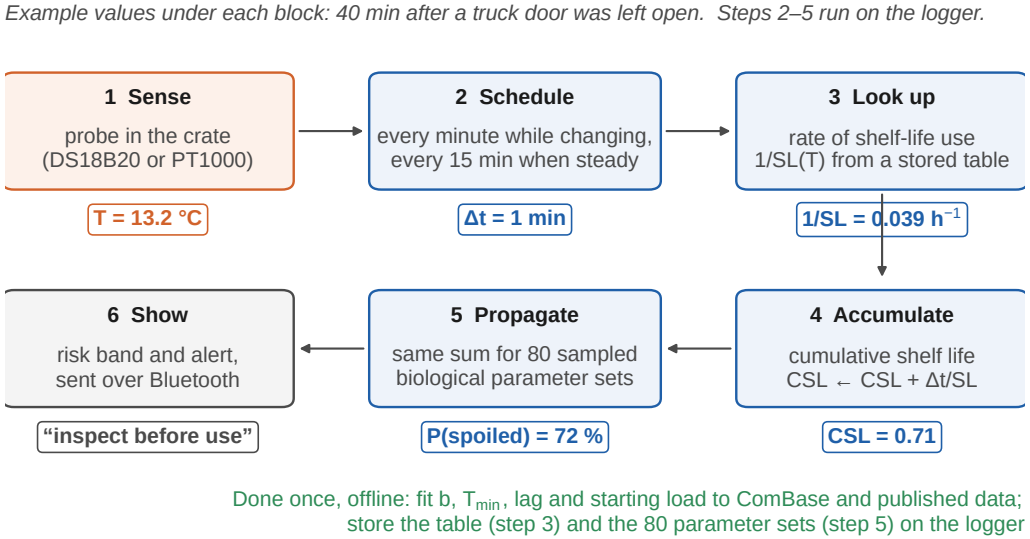


Fig. 7 Processing on the logger, with example values 40 min after a truck door was left open. Steps 2–5 run on the device; parameter fitting and the shelf-life table are prepared offline

sensor quality and sampling interval matter little; and a Monte Carlo probability is far better calibrated than an alarm. Direct measurement of biology and of growth under fluctuating temperature, with new plate counts, is the next step.

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Data and code availability

Code, 14 automated tests, the parameter file with the source of every value, all simulation results and the compiled ComBase milk records are available at https://github.com/CopyNinja07-07/dairyguard; `make all` reproduces every table and figure from fixed random seeds, and every number in this paper is written from the result files by a script. Numerical work uses NumPy, SciPy and Matplotlib [48–50]. Parts of the simulation code were developed with an AI coding assistant; all of it is covered by the automated tests and was checked by the author. ComBase records remain subject to the ComBase terms of use.

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